Topic Proposal

A Quorum-Based Distributed Security Model for Double-Spend Prevention in Hash-Chain IoT Micropayment Systems

Mustafa Can Çalışkan
Student ID: 504251544
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The topic of the term project is: **A Quorum-Based Distributed Security Model for Double-Spend Prevention in Hash-Chain IoT Micropayment Systems.**

In IoT financial systems, devices may need to make small and frequent payments. Hash-chain based micropayments are lightweight and suitable for such environments, but many of these systems assume centralized validation. On the other side, distributed payment systems are usually more complex and heavy. In literature, there is not many simple models that combine lightweight hash-chain approach with a minimal distributed quorum-based validation for double-spend prevention.

I selected this topic because I am currently studying distributed systems in my master program, and I want to combine distributed system principles with financial security problems. My aim is to design a distributed security model, without making the system too complex, and analyze how double-spending can be prevented in a structured way. Details of the model and evaluation will be clarified during the research process.